Developing a grammar to define interactive graphics with application to complex model diagnostics

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Abstract

Declaration

I hereby declare that this thesis contains no material which has been accepted for the award of any other degree or diploma at any university or equivalent institution and that, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person, except where due reference is made in the text of the thesis.

This thesis includes one original papers published in a peer reviewed journal and four unpublished papers. The core theme of the thesis is “interactive and dynamic graphics for studying non-linear dimension reduction models of high-dimensional data”. The ideas, development and writing up of all the papers in the thesis were the principal responsibility of myself, the student, working within the Department of Econometrics and Business Statistics under the supervision of Professor Dianne Cook, Dr Paul Harrison, Dr Michael Lydeamore, and Dr Thiyanga S. Talagala (University of Sri Jayewardenepura).

The inclusion of co-authors reflects the fact that the work came from active collaboration between researchers and acknowledges input into team-based research. In the case of Chapter 2, my contribution to the work involved the following:

Chapter	**Publication title**	**Status**	**Student contribution**	**Co-authors contribution**	**Coauthors are Monash students**
2		Published in the Journal of Computational and Graphical Statistics	80% Concept, Analysis, Software, Writing		No

Chapters 3, *k*, and Chapter 4, **, are planned for submission to peer-reviewed journals.

To ensure the clarity and coherence of the written content, artificial intelligence tools were employed to assist in smoothing and refining the language throughout the thesis.

I have not renumbered sections of submitted or published papers in order to generate a consistent presentation within the thesis.

Student name: Piyadi Gamage Jayani Lakshika

Student signature:

Date: 15th January 2026

I hereby certify that the above declaration correctly reflects the nature and extent of the student's and co-authors' contributions to this work. In instances where I am not the responsible author I have consulted with the responsible author to agree on the respective contributions of the authors.

Main Supervisor name: Dianne Cook

Main Supervisor signature:

Date: 15th January 2026

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Chapter 1

Introduction

Historically, 2D representations of high-dimensional data have been computed using multidimensional scaling (MDS) (Borg and Groenen 2005), which includes principal components analysis (PCA) (Jolliffe 2011) as a special case. The 2D representation can be viewed as a layout of points in 2D produced by an embedding procedure that maps the data from high dimensions. In MDS, the 2D layout is constructed by minimizing a stress function that differences distances between points in high-dimensions with potential distances between points in 2D. Various formulations of the stress function result in non-metric scaling (Saeed et al. 2018) and isomap (Silva and Tenenbaum 2002). Challenges in working with high-dimensional data, including visualization, are outlined in Johnstone and Titterington (2009).

Non-linear dimension reduction (NLDR) is popular for making a convenient low-dimensional representation of high-dimensional data by applying non-linear transformation and all designed to better capture specific structures potentially existing in high-dimensions. Here we focus on five currently popular techniques, t-distributed stochastic neighbor embedding (tSNE) (Maaten and Hinton 2008), uniform manifold approximation and projection (UMAP) (McInnes and Healy 2018), potential of heat-diffusion for affinity-based trajectory embedding (PHATE) algorithm (Moon et al. 2019), large-scale dimensionality reduction Using triplets (TriMAP) (Amid and Warmuth 2022), and pairwise controlled manifold approximation (PaCMAP) (Wang et al. 2021). tNSE and UMAP can be considered to produce the 2D minimizing the divergence between two distributions, where the distributions are modeling the inter-point distances. PHATE, TriMAP and PaCMAP are examples of diffusion processes (Coifman et al. 2005) spreading to capture geometric shapes, that include both global and local structure.

The various layouts created by Figure 1.1 demonstrate the different outcomes that can result from

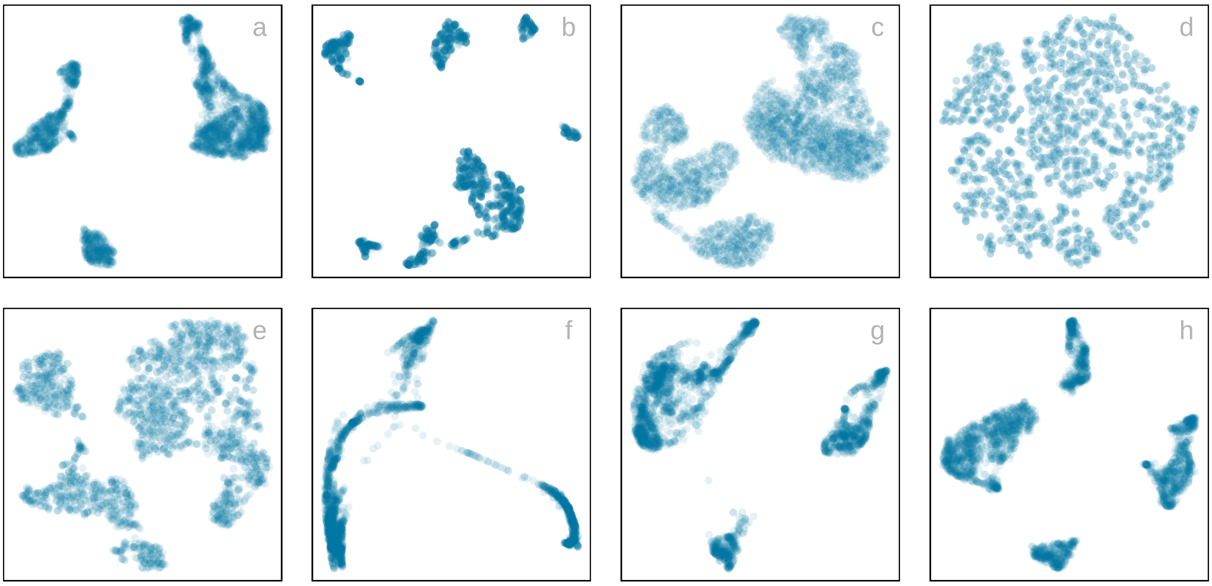

Figure 1.1: *Eight different NLDR representations of the same data. Different techniques and different parameter choices are used. Researchers may have seen any of these in their analysis of this data, depending on their choice of method, or typical parameter choice. Would they make different decisions downstream in the analysis depending on which version seen? Which is the most accurate representation of the structure in high dimensions?*

the choices of methods and parameters, as well as the random seed used to start the calculation. Key structures interpreted from these views suggest: (1) highly **separated clusters** (a, b, e, g, h) with the number ranging from 3-6; (2) **stringy branches** (f), and (3) **barely separated clusters** (c, d) which would **contradict** the other representations.

These variations occur due to the different perspectives on the distances between data points provided by the methods and parameter choices.

Another way to visualize high-dimensional data is through linear projections. PCA is the traditional method, resulting in a new set of variables that are linear combinations of the original variables. Tours, as defined by Lee et al. (2021), expand on this concept by producing dynamic linear projection visualizations that offer views of the data from all angles. Lee et al. (2021) provides an overview of the key advancements in tours. Numerous tour algorithms are available, with many included in the R package `tourr` (Wickham et al. 2011), and versions that offer enhanced interactivity in `langevitour` (Harrison 2023) and `detourr` (Hart and Wang 2022). Linear projections offer a reliable way to view high-dimensional data as they do not distort the space, making them more accurate representations of the data structure. However, linear projections can become cluttered, and global patterns can obscure local structure. The simple act of projecting data from high-dimensional spaces leads to piling (Laa et al. 2022), where data concentrates in the center of the projections.

NLDR is designed to escape these issues, to exaggerate structure so that it can be observed. However, NLDR can hallucinate wildly, to suggest patterns that are not actually present in the data. The research project addresses this significant issue by proposing methods to understand better whether the NDLR representations are reliable or hallucinations. These methods are designed to enable a more profound understanding of the quality and validity of NLDR-derived low-dimensional representations. By incorporating rigorous evaluation techniques, the project seeks to identify instances where the NLDR models genuinely capture the essential structures present in the data and where they might introduce artifacts or distortions, leading to potentially misleading representations. This meticulous examination and validation process contribute to enhancing the robustness and credibility of NLDR techniques in high-dimensional data analysis.

1.1 Thesis Outline

The rest of the thesis is organized as follows:

Chapter 2 introduces an algorithm to assess the NLDR and decide on which, if any, is the most reasonable representation of the structure(s) present in high-dimensional data. We create a model starting with NLDR layout that is then used to display as a wireframe in high dimensions.

Chapter 3 presents the implementation of the work is available as an R package, named `quollr`, an acronym to "**qu**estioning how a high-dimensional **o**bject **l**ooks in **l**ow-dimensions using **r**" (Gamage et al. n.d.a). This package also contains a function for performing hexagonal binning using a new approach, for saving langevitour results with a specific projection, and link plots to understand the quirks that occur with different NLDR techniques.

Chapter 4 provides empirical evidence on how viewers recognize structure differently when using NLDR layouts versus the tour view, particularly with varying distances between clusters. The findings will help clarify common mistakes made when selecting and reporting structures based on NLDR layouts.

Chapter 5 proposes the R package, `cardinalR` (Gamage et al. n.d.b), which includes functions to generate a large variety of structures in high dimensions along with some already generated examples.

Chapter 6 introduces `ViMRoR` (Visualize Most Reasonable Representations), a Shiny web application designed to support automated selection and diagnostic evaluation of NLDR methods.

Chapter 7 concludes the thesis, summarises the contribution of the work, and discusses some future plans.

Chapter 2

Visualising how non-linear dimension reduction warps your data

Chapter 3

quollr: An R Package for Visualizing $2-D$ Models from Non-linear Dimension Reductions in High Dimensional Space

Chapter 4

A user study to explore perception and misperception in NLDR representations

Chapter 5

cardinalR: Generating interesting high-dimensional data structures

Chapter 6

Development of an interactive and dynamic graphics system for NLDR users

6.1 Availability

Chapter 7

Conclusion and future plans

The five pieces of work assembled in this thesis share a common theme of advancing Non-linear dimension reduction (NLDR) diagnostics, with a focus on improving the NLDR methods, visual inference, and developing innovative solutions to automate diagnostic processes.

7.1 Contributions

The primary contributions of this research are fivefold. Firstly, we develop an algorithm. Secondly, we develop a package. Next, package for high-dimensional data structures. Then, a user study to explore perception and misperception in NLDR representations. Lastly, we share a user-focused R package and Shiny app, making the automated diagnostic tools accessible to a broad range of analysts and practitioners.

The aforementioned R package (and its dependency) is available on CRAN with the latest development versions in the links below:

- `quollr` (https://github.com/jayanilakshika/quollr), and
- `cardinalR` (https://github.com/jayanilakshika/cardinalR).

The Shiny app for `quollr` is available at one of the mirror sites listed at https://vimror.netlify.app/ with the source code available at https://github.com/jayanilakshika/ViMRoR.

Principles of transparency and reproducible research have guided the work with all materials related to the thesis at https://github.com/JayaniLakshika/Monash_PhD_thesis. The thesis is written using Quarto (?) and is available online at https://jayani-lakshika-phd-thesis.netlify.app.

The R packages used throughout the thesis include `tidyverse` (**?**), `lmtest` (**?**), `kableExtra` (**?**), `patchwork` (**?**), `glue` (**?**), `ggpcp` (**?**), `here` (**?**), and `knitr` (**?**). (Need to finalize at the end of writing)

7.2 Future work

There are several directions that this work can be developed.

- Scagnostics to evaluate NLDR
- Prediction with model built
- More diagnostics for NLDR (diadem app)
- Visualizations to validate experiment design/results
- Investigate perception and misperception happening with background noise, number of clusters, noise dimensions, sample size, seed.

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Appendix A

Appendix to “Visualising how non-linear dimension reduction warps your data”

Appendix B

Appendix to “quollr”

Appendix C

Appendix to “A user study to explore perception and misperception in NLDR representations”

Appendix D

Appendix to “cardinalR”

Appendix E

Appendix to “Development of an interactive and dynamic graphics system for NLDR users”